OS LAB 06

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Roll No 24K-0535

Task 1 (Screenshot)

```
student@student:~/24k-0535$ echo " Parent send integer instead of string and child receive a
nd print the same number"
 Parent send integer instead of string and child receive and print the same number
student@student:~/24k-0535$ echo " In the 2nd case if i change the buffer size to 10 so only
 10 characters will be stored and remaining characters are ignored."
 In the 2nd case if i change the buffer size to 10 so only 10 characters will be stored and
remaining characters are ignored.
student@student:~/24k-0535$
```

Task 2 (Screenshot)

```
student@student:~/24k-0535$ gedit Q2.c
student@student:~/24k-0535$ gcc Q2.c -o Q2sol
student@student:~/24k-0535$ ./Q2sol
Enter a string: Hello World
Reversed String: dlroW olleH
student@student:~/24k-0535$
```

Open

```c
#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main(){

    int fd[2];
    char str[50];
    char rev[50];
    int pid;

    pipe(fd);
    pid = fork();

    if(pid > 0){
        // Parent
        printf("Enter a string: ");
        fgets(str, 50, stdin);

        write(fd[1], str, strlen(str)+1);
    }

    else{
        // Child
        read(fd[0], str, sizeof(str));

        int len = strlen(str);

        for(int i=0;i<len-1;i++){
            rev[i] = str[len-2-i];
        }

        rev[len-1] = '\0';

        printf("Reversed String: %s\n", rev);
    }

    return 0;
}
```

Task 3 (Screenshot)

writer.c ~/24k-0535

```c
#include <stdio.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <unistd.h>

#define FIFO "/tmp/myfifo"

int main(){

    int fd;
    int num1, num2, choice;
    int data[3];

    mkfifo(FIFO,0666);

    printf("Enter two numbers: ");
    scanf("%d %d",&num1,&num2);

    printf("Enter choice (1=Sum , 2=Product): ");
    scanf("%d",&choice);

    data[0]=num1;
    data[1]=num2;
    data[2]=choice;

    fd=open(FIFO,O_WRONLY);
    write(fd,data,sizeof(data));

    close(fd);

    return 0;
}
```

rea... ~/24k...

```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

#define FIFO "/tmp/myfifo"

int main(){

    int fd;
    int data[3];
    int result;

    fd=open(FIFO,O_RDONLY);
    read(fd,data,sizeof(data));

    if(data[2]==1)
        result=data[0]+data[1];

    else if(data[2]==2)
        result=data[0]*data[1];

    printf("Result: %d\n",result);

    close(fd);

    return 0;
}
```

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student@student: ~/24...

```
student@student:~/24k-0535$ ./writer
Enter two numbers: 2 4
Enter choice (1=Sum , 2=Product): 2
student@student:~/24k-0535$
```

student@student: ~/24k-0...

```
student@student:~/24k-0535$ gedit writer.c
student@student:~/24k-0535$ gedit reader.c
student@student:~/24k-0535$ gcc writer.c -o writer
student@student:~/24k-0535$ gcc reader.c -o reader
student@student:~/24k-0535$ ./reader
Result: 8
student@student:~/24k-0535$
```